

TP8: Data Processing and Distributed

Exercise 1: Understanding Distributed Processing

Explain why distributed processing is needed in big data systems.

Tasks

1. Define distributed data processing.
2. Explain the following concepts:
 - Volume
 - Speed
 - Reliability
 - Cost
3. Give one real-world example where distributed processing is useful.

Exercise 2: Distributed Sales Data Analysis using PySpark

Objectives:

- Load large CSV data
- Perform distributed analysis
- Apply grouping and aggregation
- Compare Spark with normal Python processing

Tasks

1. Load dataset
2. Clean missing values
3. Compute:
 - Total sales
 - Top-selling store
 - Average sales
4. Visualize results
5. Explain how Spark distributes